

SECI2243-03
PROBABILITY & STATISTICAL DATA ANALYSIS

SEMESTER 2, SESSION 2025/2026

PROJECT 1 REPORT

A DESCRIPTIVE STATISTICAL ANALYSIS OF STUDENTS' BODY MASS INDEX (BMI)

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GROUP 1

GROUP MEMBERS:

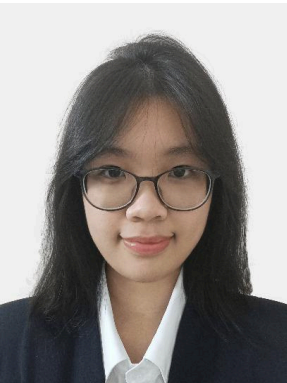
			
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1.0 INTRODUCTION

The project titled “A Descriptive Statistical Analysis of Students’ Body Mass Index (BMI)” is conducted to examine the distribution of BMI and evaluate the overall health condition among university’s students by using descriptive statistical methods.

In recent years, significant changes in the lifestyle of university students have been observed due to stress level, dietary habits, genetics, lack of physical activity and university supports. These factors have a direct impact on students’ physical health, particularly on their body weight and BMI levels. Therefore, BMI is widely used as an indicator to assess whether an individual has a healthy body weight relative to their height.

BMI is commonly used to classify individuals into several categories, namely underweight, normal weight, overweight, and obese. Each category is associated with different levels of health risk. By analyzing the distribution of BMI among students, insights into their general health condition can be obtained, and potential health concerns can be identified. In this study, all data needed are collected through a questionnaire distributed among university students. The collected data are analyzed by using R programming. Descriptive statistical tools such as frequency distribution, bar charts, histograms, and measures of central tendency are applied to interpret the data.

It is expected that the majority of students will fall within the normal BMI category. However, due to modern lifestyle habits, a certain proportion of students may be classified as underweight or overweight. The findings of this study are expected to provide a better understanding of students’ health conditions and highlight the importance of maintaining a balanced lifestyle.

2.0 DATA COLLECTION

The methodology that was used in this project to collect data is through conducting a survey via Google Form. The survey consists of several parts, which are the personal information of the respondent, the weight and height of the respondent, and the frequency of exercise of the respondent. Some of the questions in the Google Form will be shown in the appendix section at the end of the project.

NO	QUESTION	ANSWER	LEVEL OF MEASURE
1	Gender	• Male • Female	Nominal
2	Age	• Short-answer text (whole number)	Ratio
3	Faculty	• Faculty of Civil Engineering (FKA) • Faculty of Mechanical Engineering (FKM) • Faculty of Electrical Engineering (FKE) • Faculty of Chemical & Energy Engineering (FKT) • Faculty of Computing (FC) • Faculty of Science (FS) • Faculty of Social Sciences & Humanities (FSSK) • Faculty of Management (FP) • Faculty of Educational Sciences & Technology (FEST) • Other: _____	Nominal
4	Height (m)	• Short-answer text (integer number<2)	Ratio
5	Weight (kg)	• Short-answer text (integer)	Ratio
6	BMI (kg/m ²)	• Short-answer text (floating number)	Ratio
7	BMI category	• Underweight (<18.5) • Normal (18.5 - 24.9) • Overweight (25 - 29.9) • Obese (≥ 30)	Ordinal
8	Which physical activity best describes your current routine?	• Sedentary (little to no exercise) • Lightly Active (light	Ordinal

		exercise/sports 1-3 days/week) • Moderately Active (moderate exercise/sports 3-5 days/week) • Very Active (hard exercise/sports 6-7 days/week) • Extremely Active (hard daily exercise/sports and a physical job)	
9	On average, how many days per week do you consume fast food or high-calorie processed snacks?	• 0 days • 1 - 2 days • 3 - 4 days • 5+ days	Ordinal
10	How would you rate your level of awareness regarding the health risks associated with different BMI categories?	• 1(Very unaware) - 5(Very aware)	Ordinal
11	Which factor do you believe has the largest influence on your current BMI? 1. Dietary Habits 2. Physical Activity Level 3. Genetics/Heredity 4. Stress Level/Sleep Quality	• No influence • Minor influence • Moderate influence • Major influence	Ordinal
12	Do you actively try to maintain a healthy weight or BMI?	• Yes, consistently • Yes, occasionally • No, not currently	Nominal
13	Please indicate your agreement with the following statements regarding university support for healthy living: 1. The university offers adequate healthy food choices on campus. 2. The university	• Strongly Disagree • Disagree • Neutral • Agree • Strongly Agree	Ordinal

	provides sufficient sports/recreational facilities. 3. There is enough promotion of health and wellness programs by the university.		
14	Do you feel that university life (e.g., academic stress, food options on campus) negatively impacts your ability to maintain a healthy BMI?	• Yes, significantly • Yes, slightly • No, not really • Not applicable	Nominal
15	On a scale of 1 (Low) to 5 (High), how motivated are you to make lifestyle changes to achieve a healthier BMI?	• 1 (Very Low Motivation) - 5 (Very High Motivation)	Ordinal
16	If you are currently trying to maintain a healthy weight/BMI, what is your primary method?	☐ Regular exercise ☐ Calorie counting/Diet monitoring ☐ Increasing water intake/Hydration ☐ Prioritizing sleep ☐ Stress management ☐ A combination of the above ☐ Other	Nominal

3.0 DATA ANALYSIS

3.1 Respondents' Biography

3.1.1 Respondents' Age

AGE	FREQUENCY
18	1
19	6
20	25
21	9
22	7
23	5
24	4
25	3
TOTAL	60

Table 3.1.1.1: Frequency Distribution of Age of Respondent

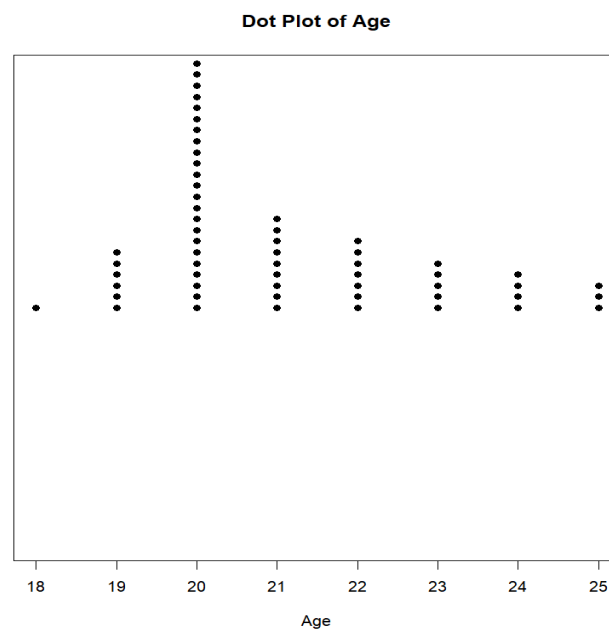


Figure 3.1.1.1: Dot Plot of Age of Respondent

Measure of Central Tendency	Value
Mean	21.01667
Median	20
Mode	20

Table 3.1.1.2: Measure of Central Tendency of Age of Respondents

As seen in Table 3.1.1.1 and Figure 3.1.1.1, the age distribution of respondents that have answered our questionnaire are around the range of the age 18 to 25. This research is conducted around the students in UTM campus and other universities campus, most of the respondents are of age 20, with frequency 25. Students of age 18, 24 and 25 are less than 5. Students of age 19 come close with the frequency of six, and there are five 23-years-old respondents. There are also nine 21-years-old and seven 22-years-old respondents. Additionally, it can be seen from Table 3.1.1.2 that the data we gather will mostly be from respondents of age 20, as our median and mode both are 20 and its mean being 21.01667.

3.1.2 Respondents' Gender

GENDER	FREQUENCY	PERCENTAGE (%)
MALE	30	50
FEMALE	30	50
TOTAL	60	100

Table 3.1.2.1: Frequency and Percentage of Gender of Respondent

Gender Distribution

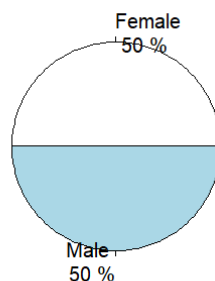


Figure 3.1.2.1: Pie Chart of Gender Distribution

As indicated in Table 3.1.2.1 and Figure 3.1.2.1, which are the frequency table and pie chart of respondents' gender, the frequency of male and female is 30. As a result, both will have the same relative frequency, 0.5, as well as the same proportion, 50%. There are 60 respondents of each gender. In addition, the pie chart of respondents' gender illustrates that white means female and blue means male.

3.1.3 Respondents' Faculty

FACULTY	FREQUENCY
Faculty of Built Environment & Surveying (FABU)	1
Faculty of Chemical & Energy Engineering (FKT)	2
Faculty of Civil Engineering (FKA)	1
Faculty of Computing (FC)	35
Faculty of Electrical Engineering (FKE)	8
Faculty of Management (FP)	1
Faculty of Mechanical Engineering (FKM)	1
Faculty of Science (FS)	8
Faculty of Social Sciences & Humanities (FSSK)	1
USM School of Communications	2
TOTAL	60

Table 3.1.3.1: Frequency Distribution of Faculty of Respondents

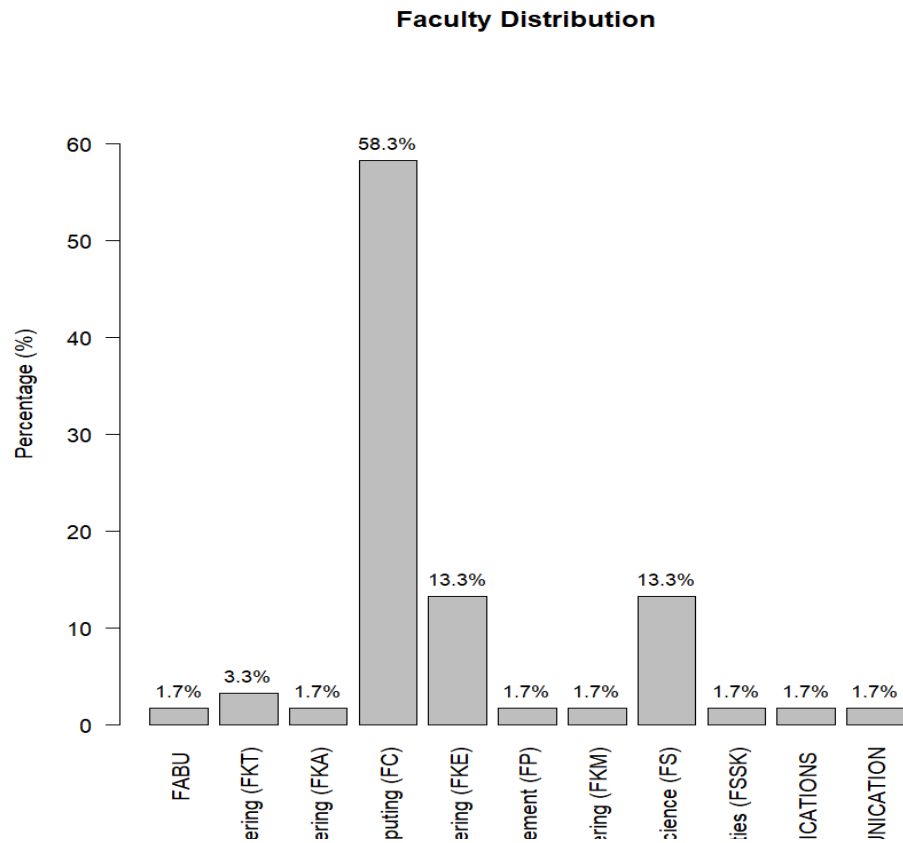


Figure 3.1.3.1: Frequency Distribution of Faculty Distribution

As stated in Table 3.1.3.1 and Figure 3.1.3.1, the respondent's faculty in UTM and other campus universities which is USM campus. The faculty with the most respondents is Faculty of Computer Science (FC) with 35 respondents. Following that, Fabu, FP, FKA, FKM, FSSK, USM COMMUNICATIONS, both have only one respondent. FKA and FC both have 8 respondents while FKT have 2 respondents. We can conclude that 35 out of a total 60 respondents are FC as well as the mode of Faculty.

3.2 Respondents' BMI Calculation

3.2.1 Respondents' Height

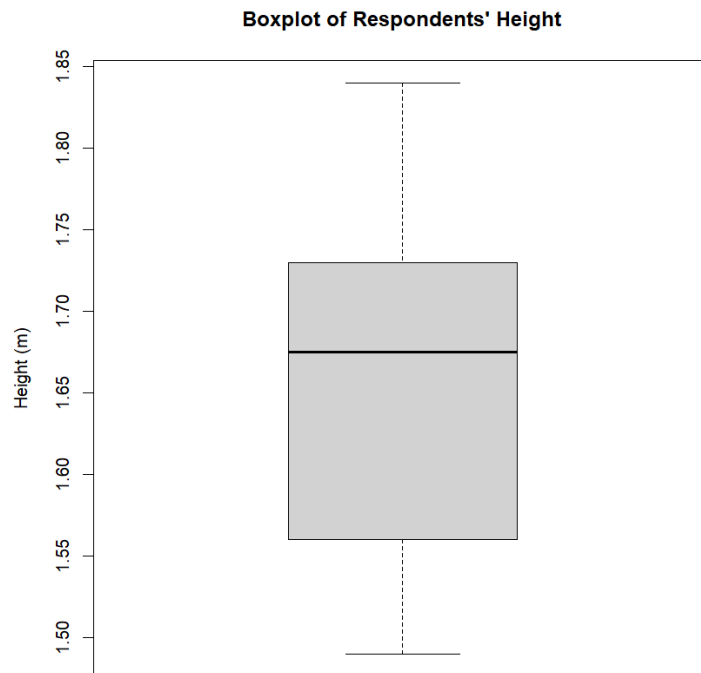


Figure 3.2.1.1: Boxplot of Respondents' Height

Measure of Central Tendency and Position	Value
Mean	1.657
Mode	1.700
Median	1.675
1st Quartile	1.565
3rd Quartile	1.730
Minimum Value	1.490
Maximum Value	1.840

Table 3.2.1.1: Measure of Central Tendency and Position of Respondents' Height

Based on the figure 3.2.1.1, show that 60 respondents' height with maximum, minimum, median and its' quartile which is 3rd and 1st quartile. The figure shows that the min height is less than 1.5m which is 1.490m while the max height is higher than 1.8m which is 1.840m. The table 3.2.1.1 also shows that the mean of respondents height is 1.657m and the most of the height is 1.7m. The median of the respondents' height is 1.675m, 1.565m and 1.730m are first quartile and third quartile of the respondents' height. We can conclude that the median height is the best measure of central tendency.

3.2.2 Respondents' Weight

WEIGHT CLASS	FREQUENCY
36 - 41	5
41 - 46	10
46 - 51	6
51 - 56	6
56 - 61	6
61 - 66	6
66 - 71	7
71 - 76	5
76 - 81	3
81 - 86	2
86 - 91	2
91 - 96	2
TOTAL	60

Table 3.2.2.1: Frequency of Group Data of Respondents' Weight

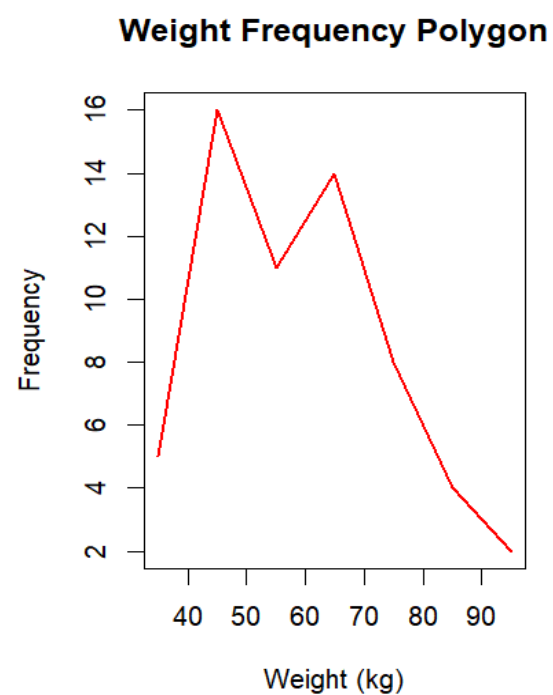


Figure 3.2.2.1: Weight Frequency Polygon

Measure of Central Tendency and Position	Value
Mean	59.41
Median	60
1st Quartile	45.75
3rd Quartile	69.25
Minimum Value	36
Maximum Value	95

Table 3.2.2.2: Measure of Central Tendency and Position

As stated in Table 3.2.2.1 show the frequency of the respondents' weight in group data. Figure 3.2.2.1 showed the respondents' weight in polygon frequency. The table 3.2.2.2 measures that the mean weight is 59.41kg. The first quartile is 45.75kg, the second quartile is 60kg and its' min and max are 36kg and 95kg.

3.2.3 Respondents BMI

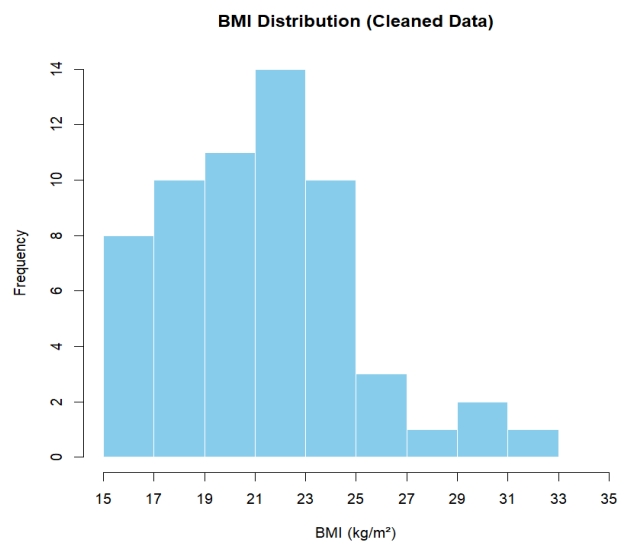


Figure 3.2.3.1: Respondents' BMI distribution

Measure of Central Tendency and Position	Value
Mean	21.36
Median	21.15
1st Quartile	18.77
3rd Quartile	23.27
Minimum Value	15.40
Maximum Value	32.87

Table 3.2.2.1: Measure of central tendency and position of respondents' BMI

As shown in Table 3.2.3.1 and Figure 3.2.3.1, the Body Mass Index (BMI) of respondents was analysed using descriptive statistics and a boxplot. The mean BMI of respondents is 21.36, while the median is 21.15, indicating that the data distribution is slightly right-skewed since the mean is slightly higher than the median. The first quartile (Q1) is 18.77 and the third quartile (Q3) is 23.27, showing that 50% of respondents have BMI values between this range. The minimum and maximum BMI values are 15.40 and 32.87 respectively, showing a relatively wide range of BMI among respondents. From the boxplot, most respondents fall within the normal BMI range, with a few higher values indicating possible outliers. Overall, it can be concluded that the respondents' BMI is generally concentrated around the normal range, with a slight positive skewness in the distribution.

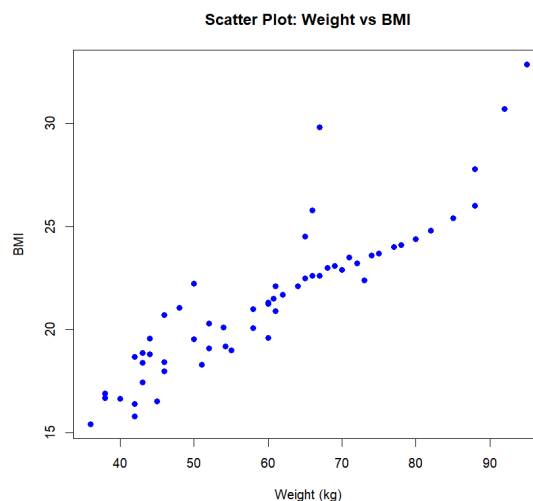


Figure 3.2.3.2: Comparison Between Weight and BMI Using Scatter Plot

Figure 3.2.3.2 shows the relationship between weight and BMI. There is a positive relationship, where higher weight tends to be associated with higher BMI. This indicates that weight is a strong contributing factor to BMI among respondents.

3.2.4 Respondents' BMI Category

BMI CATEGORY	FREQUENCY	PROPORTION
Underweight (<18.5)	13	0.2167
Normal (18.5 - 24.9)	40	0.6667
Overweight (25 - 29.9)	5	0.0833
Obese (>=30)	2	0.0333
TOTAL	60	1.0000

Table 3.2.4.1: Frequency of Respondents' BMI category

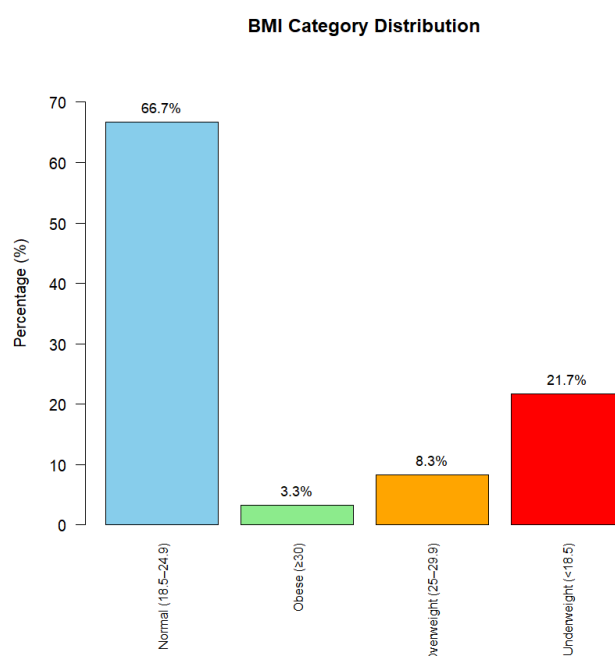


Figure 3.2.4.1: BMI Category Distribution

As shown in Table 3.2.4.1 and figure 3.2.4.1, the BMI categories of respondents were classified into underweight, normal, overweight, and obese based on WHO standards. The majority of respondents fall under the normal category with 40 respondents (66.67%). This is followed by 13 respondents (21.67%) in the underweight category. Only a small proportion of respondents are classified as overweight with 5 respondents (8.33%), while the obese category has the least number of respondents with 2 respondents (3.33%). From this distribution, it can be concluded that most respondents are within the normal BMI range, indicating a generally healthy weight status among the respondents. However, a small proportion of respondents fall into underweight and overweight/obese categories, which may require attention.

3.3 Respondents' Lifestyle and Motivation

3.3.1 Respondents' Physical Activity Routine

Which physical activity best describes your current routine?

Total responses: 60

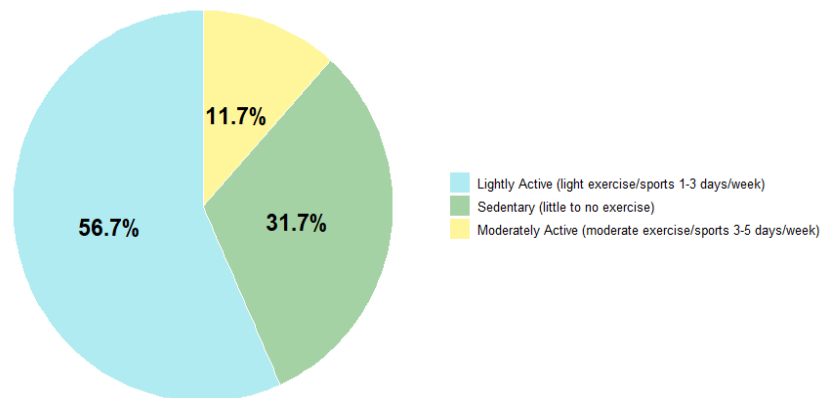


Figure 3.3.1.1: Pie Chart of Respondents' Physical Activity Routines

PHYSICAL ACTIVITY LEVEL	FREQUENCY	PROPORTION
Sedentary (little to no exercise)	19	0.3167
Lightly Active (light exercise/sports 1-3 days/week)	34	0.5667
Moderately Active (moderate exercise/sports 3-5 days/week)	7	0.1167
Very Active (hard exercise/sports 6-7 days/week)	0	0.0000
Extremely Active (hard daily exercise/sports and a physical job)	0	0.0000
TOTAL	60	1.0000

Table 3.3.1.1: Frequency of Respondents' Physical Activity Routines

Figure 3.3.1.1 demonstrates an intense concentration of people who spend their time being active but not too much. "Lightly Active" people form the most populated segment of the graph accounting for 56.7% of its area. It means that the largest proportion of the studied group of 60 respondents does exercises not more than 1–3 days in a week. A considerable gap can be observed between the percentage of active and sedentary respondents within the examined population. 88.4% of the surveyed individuals belong to the "Sedentary" and "Lightly Active" segments. Only a small number of respondents (11.7%) are able to become "Moderately Active." The absence of "Very Active" and "Extremely Active" segments on the graphic image demonstrates that for these particular people, physical activities take second place among all other obligations.

3.3.2 Respondents' Fast Food Consumption

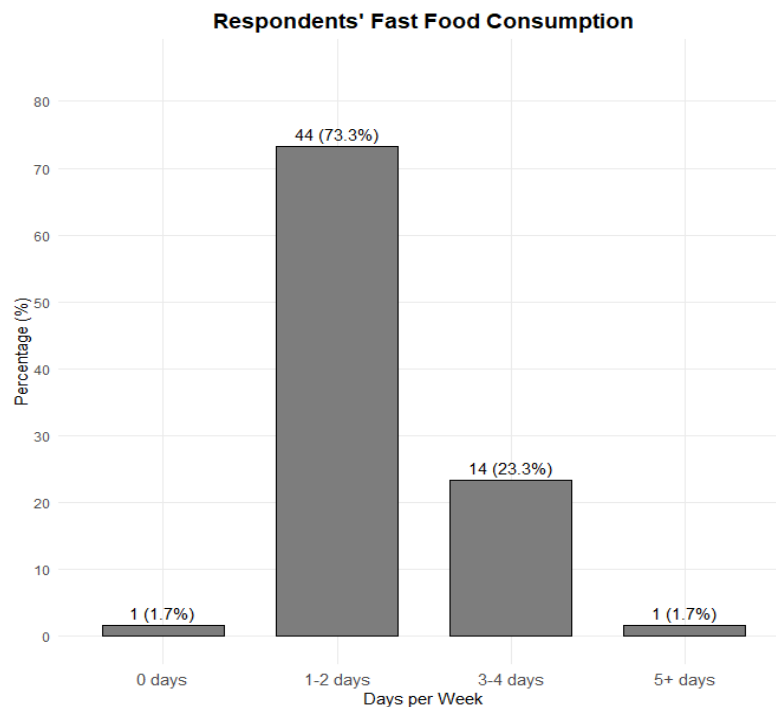


Figure 3.3.2.1: Bar Chart of Respondents' Fast Food Consumption

FAST FOOD CONSUMPTION	FREQUENCY	PROPORTION
0 days	1	0.0167
1 - 2 days	44	0.7333
3 - 4 days	14	0.2333
5+ days	1	0.0167
TOTAL	60	1.0000

Table 3.3.2.1: Frequency of Respondents' Fast Food Consumption

Figure 3.3.2.1 shows that the distribution pattern has a peak in one location, specifically on the "1-2 days." Besides that, there is an abundance of observations around the center, which shows that almost 75% of the total number of 60 people consume fast food moderately. The very small percentages in the "0 days" and "5+ days" groups (both 1.7%) show that complete abstinence from fast food and excessive consumption of fast food each day are outliers for this group. Most individuals lie within the 1-4 days bracket. It indicates that although the population might be highly motivated to change their lifestyles, fast food is part of their lifestyle but in a moderate amount. It corroborates with other findings where the people prefer focusing on "low effort" lifestyle behaviors like hydration and sleep rather than calorie counting.

3.3.3 Respondents' Awareness of BMI Health Risks

AWARENESS RATING (1-5)	FREQUENCY	PROPORTION
1 (Very Low)	1	0.0167
2 (Low)	6	0.1000
3 (Moderate)	11	0.1833
4 (High)	20	0.3333
5 (Very High)	22	0.3667
TOTAL	60	1.0000

Table 3.3.3.1: Frequency of Awareness Levels Regarding BMI Health Risks

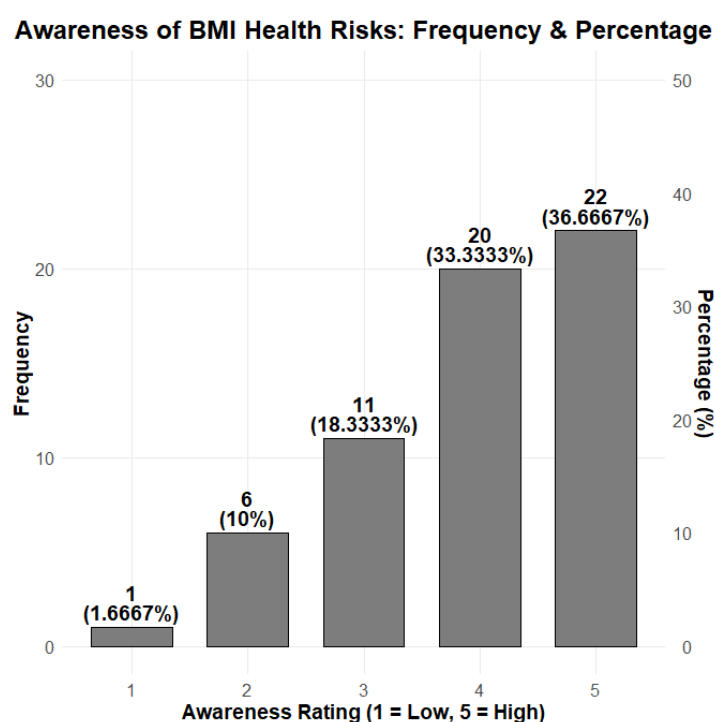


Figure 3.3.3.1: Respondents' Awareness Levels Regarding BMI Health Risks

The analysis of Figure 3.3.3.1 reveals a left-skewed distribution among the 60 respondents, with frequencies increasing alongside awareness levels. The data is heavily concentrated at the top of the scale, with the "Very High" category serving as the mode at 36.6667%. High health literacy is dominant, as 70% of participants reported "High" or "Very High" awareness, while only 1.6667% reported "Very Low" knowledge.

3.3.4 Comparison of Influencing Factors on BMI

FACTOR	NO INFLUENCE	MINOR INFLUENCE	MODERATE INFLUENCE	MAJOR INFLUENCE	TOTAL
Dietary Habits	0	1	19	40	60
Physical Activity Level	1	4	21	34	60
Genetics/Hereditary	0	11	21	28	60
Stress Level/Sleep Quality	3	6	22	29	60
TOTAL	4	22	83	131	

Table 3.3.4.1: Perceived Influence of Lifestyle and Biological Factors on BMI

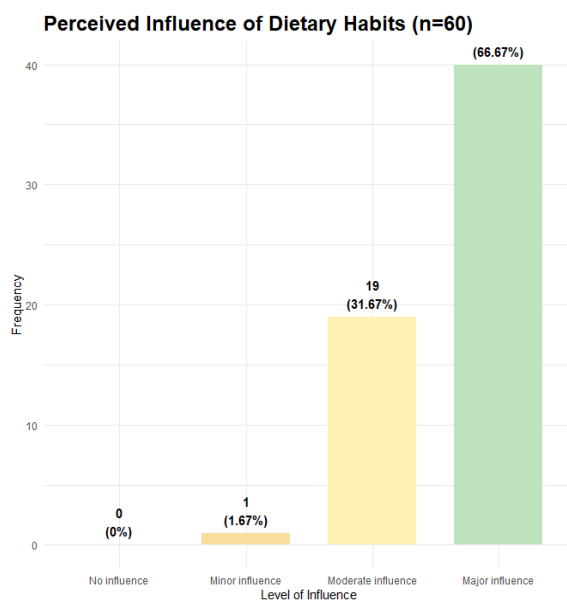


Figure 3.3.4.1: Influence of Dietary Habits on BMI

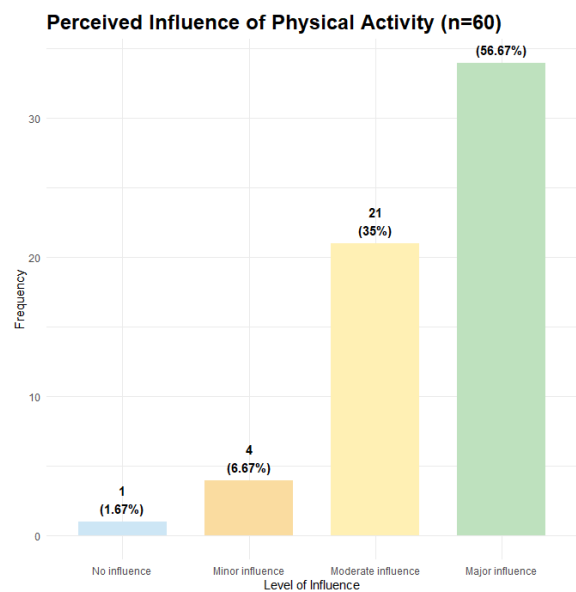


Figure 3.3.4.2: Influence of Physical Activity Level on BMI

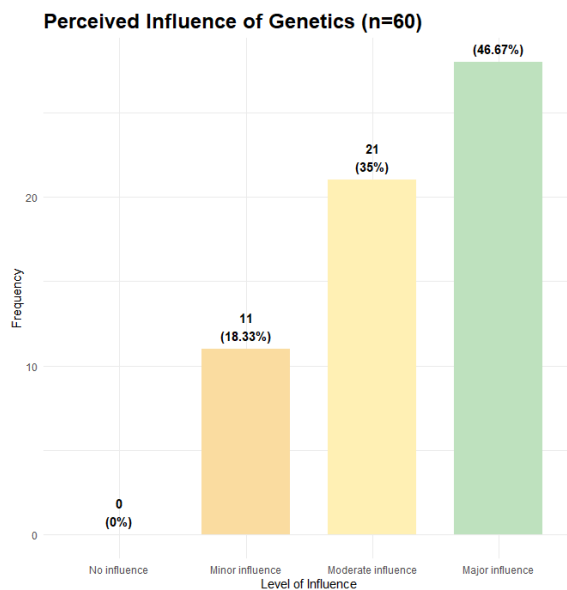


Figure 3.3.4.3: Influence of Genetics/Hereditry on BMI

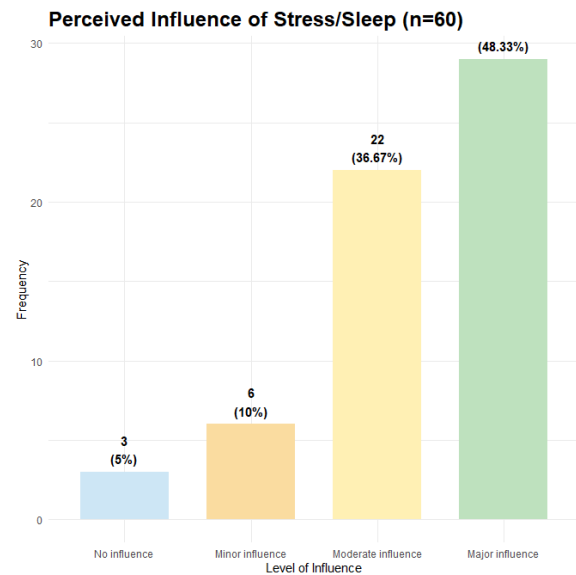


Figure 3.3.4.4: Influence of Stress Level/Sleep Quality on BMI

According to Table 3.3.4.1 and Figures 3.3.4.1, 3.3.4.2, 3.3.4.3, and 3.3.4.4 above, four main factors have been classified to influence the statistics, which are Dietary Habits, Physical Activity Level, Genetics/Hereditry, and Stress Level/Sleep Quality. In the bar charts below, green color shows the highest frequency (Major Influence) while blue color shows the lowest frequency (No Influence). It should be noted that from the 60 respondents analyzed, a clear consensus shows that lifestyle choices have the greatest impact on the development of high BMI. The factors of Dietary Habits and Physical Activity show the highest frequency under the Major Influence category. Although biological and psychological aspects, such as Genetics and Stress, are influential in determining BMI, they are distributed across various ranges.

Dietary Habits (Figure 3.3.4.1)

This criterion is of the highest overall frequency among those who have cited "Major Influence" on their BMI with 40 out of 60 subjects (66.67%) pointing to this particular criterion as the driving force behind the same. It is represented through the large green bar graph.

Physical Activity Level (Figure 3.3.4.2)

Next after diet is the importance placed on exercise, with 34 participants considering it to be a "Major Influence." Despite the fact that most participants favor major or moderate influence, exercise provides the first example of "No Influence," but only in small numbers (blue bar).

Genetics/Hereditry (Figure 3.3.4.3)

Perception of biological influence is important, but not necessarily absolute. A total of 28 students regard it as a "Major Influence," while at the same time having the most occurrences in terms of "Minor Influence" (11 students), more than any other lifestyle factors, indicating that most students can conquer biology with action.

Stress Level/Sleep Quality (Figure 3.3.4.4)

The connection between the psychological and physiological aspects of human health is made through this aspect as 29 participants consider it a "Major Influence". But still, it is also noted for having the highest number of "No Influence" answers with three participants responding to this category which is symbolized by the blue bar.

3.3.5 Respondents' Agreement

Agreement Statement	Agree	Disagree	Neutral	Strongly Agree	Strongly Disagree
Adequate Healthy Food Choices on Campus	6	27	8	8	11
Provide Sufficient Sports Recreational Facilities on Campus	16	18	5	16	5
Promotion of Health and Wellness Programs on Campus	9	22	10	12	7
TOTAL	31	67	23	36	23

Table 3.3.5.1: Respondents' Agreement Statement

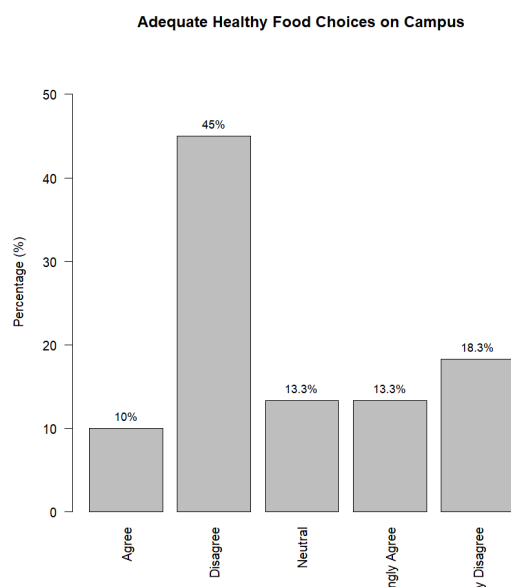


Figure 3.3.5.1: Respondent's Agreement on Adequate Healthy Food Choices on Campus

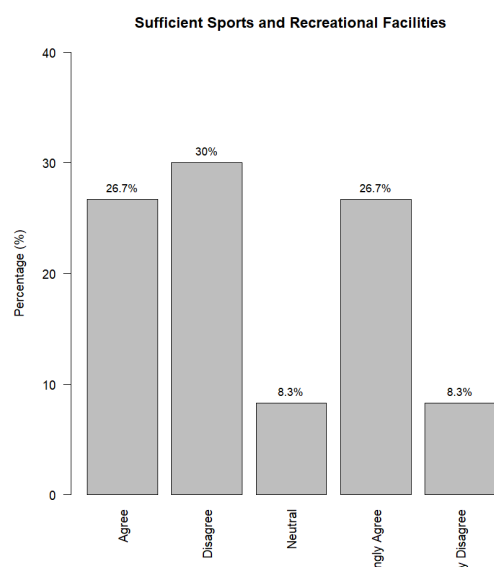


Figure 3.3.5.2: Respondents' Agreement on Providing Sufficient Sports Recreational Facilities on Campus

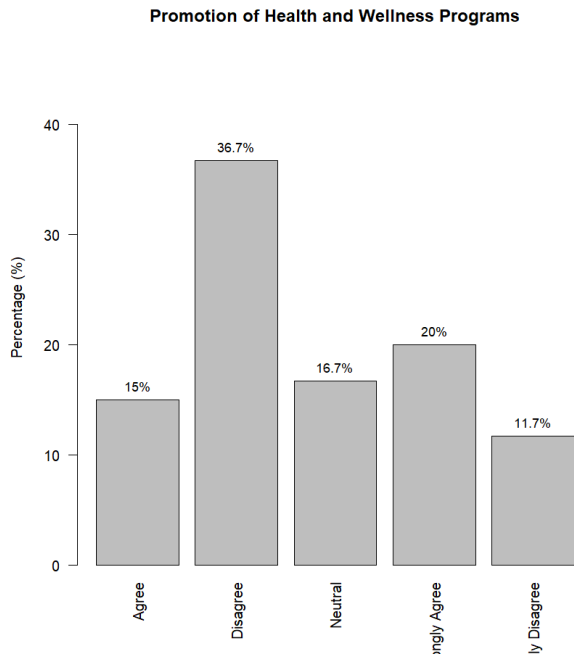


Figure 3.3.5.3: Respondents' Agreement on Promotion of Health and Wellness Programs on Campus

Based on Figure 3.3.5.1, Figure 3.3.5.2 and Figure 3.3.5.3, respondents' agreement with the statements regarding university support for healthy living that showed in the figures. The table 3.3.5.1 showed the frequency of each agreement. There are 27 respondents who disagree with adequate health food choices on campus. 18 respondents disagree with providing sufficient sports and recreational facilities on campus. 22 respondents disagree with promotion of health and wellness programs on campus. Overall, the results indicate a relatively negative perception among a considerable proportion of respondents toward university support for healthy living. Although not all students expressed disagreement, the findings suggest that improvements may be needed in the availability of healthy food options, sports facilities, and wellness promotion programs within the university.

3.3.6 Supportation Negative Impact of University Life

IMPACT LEVEL	FREQUENCY	PROPORTION
Yes, significantly	27	0.4500
Yes, slightly	25	0.4167
No, not really	6	0.1000
Not applicable	2	0.3333
TOTAL	60	1.0000

Table 3.3.6.1: Impact of University Life on Maintaining a Healthy BMI

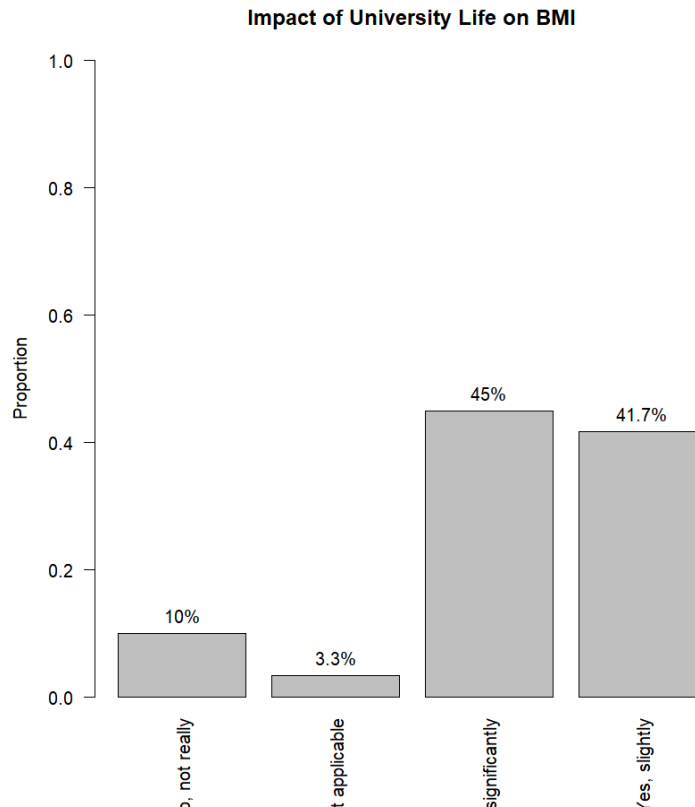


Figure 3.3.6.1: Agreement of the Respondents' on Impacting of University Life on BMI

Figure 3.3.6.1 shows that 45% of respondents significantly agree and that it negatively impacts your ability to maintain a healthy BMI due to university life like stress and food options on campus. 10% of respondents disagree about the statement. Overall, the findings suggest that a considerable proportion of students perceive university life as a contributing factor that affects their ability to maintain a healthy BMI.

3.3.7 Motivation for Lifestyle Changes

MOTIVATION LEVEL	FREQUENCY	PROPORTION
1 (Very Low)	3	0.0500
2	4	0.0667
3	14	0.2333
4	19	0.3167
5 (Very High)	20	0.3333
TOTAL	60	1.0000

Table 3.3.7.1: Frequency of Respondents' Motivation Levels

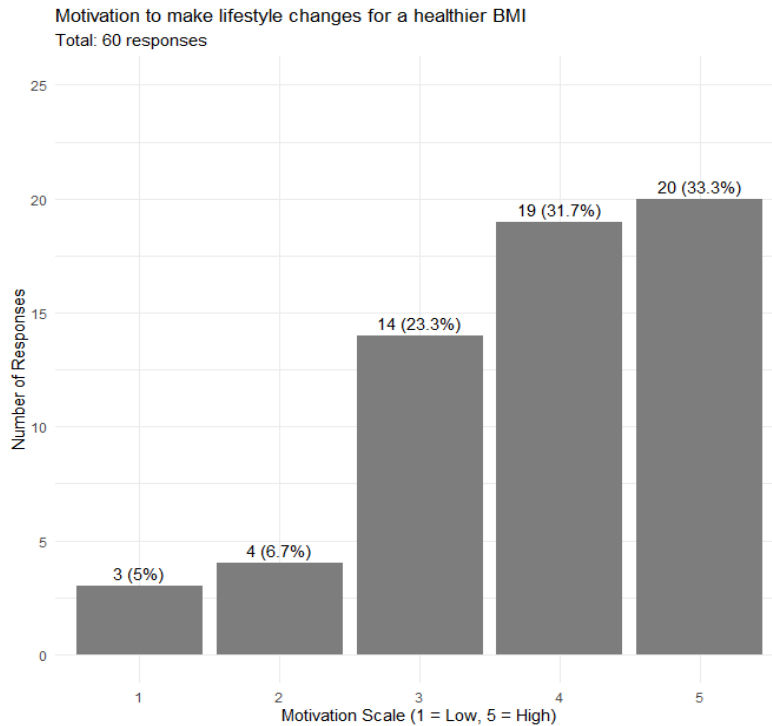


Figure 3.3.7.1: Bar Chart of Respondents' Motivation Levels

The data reveals a left-skewed (negatively skewed) distribution, as the majority of responses are concentrated at the higher end of the scale. Specifically, 65% of students reported high motivation levels of 4 or 5, while the mean motivation score of 3.8167 and a median of 4.0 further highlight this positive trend. The shape of the bar chart above indicates that as the scale increases, the frequency also rises, suggesting that most students within the faculty are highly willing to engage in health-improving behaviors. This high level of motivation is a critical finding, as it implies that wellness initiatives or dietary programs introduced by the university would likely receive strong participation and engagement from the student body.

3.3.8 Primary Method for Maintaining Healthy Weight/BMI

PRIMARY METHOD	FREQUENCY
Regular Exercise	17
Calorie counting/Diet monitoring	11
Increasing water intake/Hydration	22
Prioritizing sleep	20
Stress management	11
A combination of the above	36

Other	0
TOTAL	117

Table 3.3.8.1: Frequency of Respondents' Primary Weight Management Methods

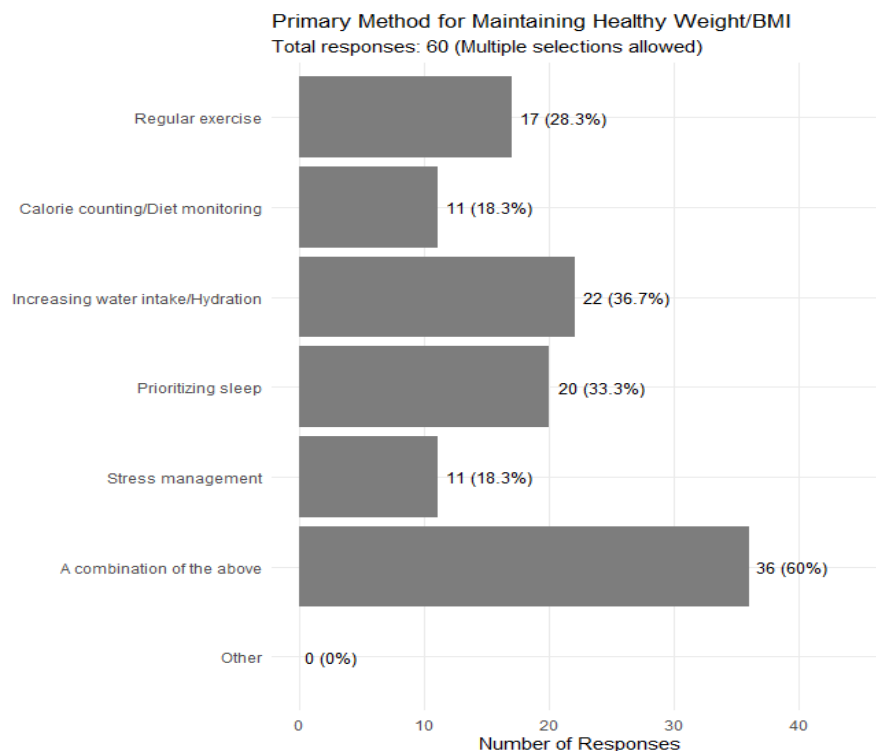


Figure 3.3.8.1: Distribution of Primary Methods for Maintaining Healthy Weight and BMI

The distribution chart in Figure 3.3.8.1 shows a distinct preference for a combined health management system. The domination of the “Combination” option (60.0%) clearly demonstrates that the majority of 60 individuals understand that a healthy BMI is not achieved through one simple measure but through a comprehensive approach to life. In the case of individual methods, the bias towards “low impact” changes is obvious. While hydration (36.7%) and sleep (33.3%) are chosen by more individuals compared to the traditional “high impact” options like exercise (28.3%) and diet monitoring (18.3%), it is evident that although the desire for change is present, most of the participants opt for habits that are easy to incorporate into a hectic routine.

4.0 CONCLUSION

Based on the data gathered using the survey conducted on 60 respondents from Universiti Teknologi Malaysia, where 30 of them are males and 30 of them are females, it can be seen that despite the majority having a healthy body condition, there are certain outliers that should be addressed. Based on the results obtained from the analysis, the Faculty of Computing (FC) is found to be the dominant demographic group, comprising 58.3% of the total population size. The age mode of the respondent is 20 years old, and this age group represents the highest number of people in the undergraduate cohort. In terms of physical measurements, the height that occurs most often lies between 1.65m and 1.75m while the weight is concentrated at 60.0 kg. Moreover, it was observed that the tallest person in the sample population stands at 1.84 m whereas the heaviest weighs 95.0 kg. Based on our findings, it was found that 66.7% of students fall within the “Normal” BMI category. On the other hand, 21.7% of respondents were classified as “Underweight”, which is significantly higher than the combined percentage of those in the “Overweight” and “Obese” categories (11.6%). The study further identified that lifestyle behaviors like dietary habits and physical activity serve as the greater determinant of BMI results. It has been observed that more than half of the respondents (56.67%) are “Lightly Active”, whereas 31.67% of the respondents are “Sedentary” routines. In terms of eating habits, the most common frequency of consumption of fast food and other calorie-rich snacks is once or twice a week (73.3%). Despite these obstacles, there exists a high level of motivation among the respondents, with 65% of the students scoring their motivation for a healthy BMI level 4 or 5. However, 86.67% of the respondents think that university life such as academic stress and the type of food available on campus affects their body mass index (BMI). As for the main strategies students adopt to keep themselves healthy and maintain a healthy weight, it can be stated that a comprehensive strategy is the most popular method. A combination of methods was selected by the highest number of respondents (60%), followed by increasing water intake/hydration (36.7%) and prioritizing sleep (33.3%). Regular exercise was utilized by 28.3% of the respondents, while calorie counting and stress management were both identified as primary methods by 18.3% of respondents. Thus, it can be said that although students are highly aware of their health status (70% claim that awareness is at levels 4 and 5), the environmental factors associated with university life constitute the biggest obstacle to achieving good health.

Several critical technical skills and analyses were gained during the course of implementing this study. In the initial stage of data collection, a Google Form survey with sixteen (16) questions was constructed as an effective medium to gather quantitative data about the BMI of UTM students. This survey was effectively distributed among four (4) localized UTM student WhatsApp groups, resulting in a variety of data collected within a week to finally achieve a sample size of 60 respondents. Descriptive statistics were proven mathematically by calculating the central tendency and the distribution of frequencies. It was found that the Median offered an excellent measure of the health status of the population. In addition, it could pinpoint the average weight of 60.0 kg despite having outliers in the data set. Technologically, using R Programming was a key factor in analyzing the collected information. It became possible to become skilled in drawing bar charts, histograms, and boxplots without much programming knowledge. The use of R would make it easier to visualize the obtained data and convert it to the needed form of percentage graphs rather

than do it manually. It is expected that the acquired proficiency in using statistical tools and collecting digital data will be highly advantageous in future workplaces. In conclusion, while students demonstrate sufficient effort to maintain physical fitness, institutional support should be provided to guarantee the availability of campus food and lessen their study load.

APPENDIX: SCREENSHOT OF GOOGLE FORM

A Descriptive Statistical Analysis of Students' Body Mass Index (BMI)

This survey is conducted as part of a project for the course SECI1143 Probability & Statistical Data Analysis. The purpose of this study is to collect data on UTM students' height and weight in order to calculate and analyze Body Mass Index (BMI).

The information collected will be used only for academic purposes and will remain confidential. Your participation is voluntary, and all responses will be kept anonymous.

Thank you for taking the time to complete this survey.

Gender *

☐ Male

☐ Female

Age *

Short answer text

Faculty *

☐ Faculty of Civil Engineering (FKA)

☐ Faculty of Mechanical Engineering (FKM)

☐ Faculty of Electrical Engineering (FKE)

☐ Faculty of Chemical & Energy Engineering (FKT)

☐ Faculty of Computing (FC)

☐ Faculty of Science (FS)

☐ Faculty of Social Sciences & Humanities (FSSK)

☐ Faculty of Management (FP)

☐ Faculty of Educational Sciences & Technology (FEST)

☐ Other: _____

Height (m) *

Short answer text

Weight (kg) *

Short answer text

BMI (kg/m²) *

Short answer text

BMI category *

- ☐ Underweight (<18.5)
- ☐ Normal (18.5–24.9)
- ☐ Overweight (25–29.9)
- ☐ Obese (≥30)

...

Which physical activity best describes your current routine? *

- ☐ Sedentary (little to no exercise)
- ☐ Lightly Active (light exercise/sports 1-3 days/week)
- ☐ Moderately Active (moderate exercise/sports 3-5 days/week)
- ☐ Very Active (hard exercise/sports 6-7 days/week)
- ☐ Extremely Active (hard daily exercise/sports and a physical job)

On average, how many days per week do you consume fast food or high-calorie processed snacks? *

- ☐ 0 days
- ☐ 1-2 days
- ☐ 3-4 days
- ☐ 5+ days

How would you rate your level of awareness regarding the health risks associated with different BMI categories? *

	1	2	3	4	5	
Very unaware	☐	☐	☐	☐	☐	Very aware

Which factor do you believe has the largest influence on your current BMI? *

	No influence	Minor influence	Moderate influence	Major influence
Dietary Habits	☐	☐	☐	☐
Physical Activity L...	☐	☐	☐	☐
Genetics/Heredity	☐	☐	☐	☐
Stress Level/Sleep...	☐	☐	☐	☐

Do you actively try to maintain a healthy weight or BMI? *

- ☐ Yes, consistently
- ☐ Yes, occasionally
- ☐ No, not currently

Do you feel that university life (e.g., academic stress, food options on campus) negatively impacts your ability to maintain a healthy BMI? *

- ☐ Yes, significantly
- ☐ Yes, slightly
- ☐ No, not really
- ☐ Not applicable

Please indicate your agreement with the following statements regarding university support for ^{*} healthy living:

	Strongly Disagr...	Disagree	Neutral	Agree	Strongly Agree
The university ...	☐	☐	☐	☐	☐
The university ...	☐	☐	☐	☐	☐
There is enough...	☐	☐	☐	☐	☐

On a scale of 1 (Low) to 5 (High), how motivated are you to make lifestyle changes to achieve ^{*} a healthier BMI?

	1	2	3	4	5	
Very Low Motivation	☐	☐	☐	☐	☐	Very High Motivation

If you are currently trying to maintain a healthy weight/BMI, what is your primary method? ^{*}

- ☐ Regular exercise
- ☐ Calorie counting/Diet monitoring
- ☐ Increasing water intake/Hydration
- ☐ Prioritizing sleep
- ☐ Stress management
- ☐ A combination of the above
- ☐ Other